Asadullah Bhuiyan

Ph.D. Candidate in Theoretical Physics | Cornell University
Ithaca, New York | ab2398@cornell.edu | asadullahbhuiyan.github.io
Google Scholar | GitHub | LinkedIn

Theoretical physicist with experience in neural-network learning theory and quantum many-body dynamics, combining analytical methods with numerical experiments.

Research interests: statistical physics of learning; interpretable descriptions of neural-network behavior; AI alignment.

EDUCATION

Cornell University Aug 2021 - Jun 2027 (expected)

Ph.D. in Theoretical Physics. Advisor: Chao-Ming Jian.
Thesis: *Random Processes in Topological Quantum Dynamics and Machine Learning*.

University of Alberta 2017 - 2021

B.Sc. in Engineering Physics, with distinction.

DOCTORAL RESEARCH

Graduate Research Assistant, Cornell University May 2022 - Present

Neural-network learning under input corruption Preprint | Code

- Initiated a study of neural networks trained on heavily corrupted inputs and recruited three collaborators.
- Used Python/PyTorch to study ensembles of finite-width networks trained with varying input corruption.
- Derived a leading-order interpretable decision rule universal across MLP architectures in the high-noise, infinite-width limit, using symmetry and linear response within neural tangent kernel theory.

Topological quantum dynamics and adaptive state preparation Paper | Code

- Developed Lie-algebraic proofs for the topological classification of monitored free-fermion dynamics; developed existence proofs for Gaussian measurements within each topological class.
- Designed and analyzed a measurement-feedback protocol for preparing disordered ensembles of free-fermion topological states.
- Developed an effective description of the dynamics and used covariance methods to determine the relaxation rate.
- Simulated the protocol's topological phase transitions using Python/PyTorch.
- Ongoing follow-up work examines critical and chiral properties of topological domain-wall modes prepared by our measurement-feedback protocol.

Stochastic bosonic Gaussian dynamics

- Proved an exponential upper bound on mean second Rényi entropy in an all-to-all bosonic Gaussian circuit with Haar-random passive unitary gates and single-mode heterodyne measurements.
- Constructed a replica spin-chain theory describing entropy dynamics in a 1+1D stochastic, monitored Gaussian bosonic circuit.
- Simulated entanglement entropy dynamics in Gaussian bosonic circuits via covariance methods and Python.

Earlier Research and Industry Experience

Undergraduate Research Assistant, University of Alberta May 2018 - Aug 2021

Developed an iterative MATLAB Green's-function solver to study superconducting phase transitions in the 2D Hubbard-Holstein model using dynamical mean-field theory. Other projects include: Floquet quantum systems, quantum Hall physics, and experimental biophysics with microtubules.

Product Engineering Intern, Microchip Technology Inc. Sep 2019 – Dec 2020

Designed hardware reliability tests for microchips under extreme electrical and thermal stress, built a Python pipeline to automate testing and analysis, and reported findings to leadership.

Publications

1. J. Tahmassebpur, A. Bhuiyan, H. Kim, and O. Lesser. Learning from almost nothing: How neural networks survive heavy input corruption. arXiv:2606.11319. Accepted by *Physical Review Research* (2026).
2. A. Bhuiyan, H. Pan, and C.-M. Jian. Free-Fermion Dynamics with Measurements: Topological Classification and Adaptive Preparation of Topological States. *Physical Review Research* **8**, 023147 (2026).
3. R. Boyack, A. Bhuiyan, A. Su, and F. Marsiglio. The bound-state solutions of the one-dimensional pseudoharmonic oscillator. *Journal of Mathematical Chemistry* **61**, 242-276 (2023).
4. A. Bhuiyan and F. Marsiglio. Edge Localized Schrödinger Cat States in Finite Lattices via Periodic Driving. *Physical Review B* **102**, 245302 (2020).
5. A. Bhuiyan and F. Marsiglio. Landau Levels, Edge States, and Gauge Choice in 2D Quantum Dots. *American Journal of Physics* **88**, 986-1005 (2020).
6. A. P. Kalra, S. D. Patel, A. F. Bhuiyan, J. Preto, K. G. Scheuer, U. Mohammed, J. D. Lewis, V. Rezania, K. Shankar, and J. A. Tuszynski. Investigation of the Electrical Properties of Microtubule Ensembles under Cell-Like Conditions. *Nanomaterials* **10**, 265 (2020).

Talks

Quantum Nano Seminar, Dartmouth College 11 Nov 2025

Fermionic Monitored Dynamical Systems: Topological Classification & Adaptive Circuit Constructions. Invited research seminar.

APS March Meeting, Anaheim 19 Mar 2025

Topological Modes in Monitored Dynamics.

APS March Meeting, Las Vegas 6 Mar 2023

Free Bosons Purify Faster than Free Fermions in 0D.

Honors and Awards

Cornell Graduate Fellowship 2025

One-year fellowship supporting graduate research.

NSERC Postgraduate Scholarship - Doctoral (PGS-D) 2022

Three-year scholarship supporting graduate research.

Advanced Training

Prospects in Theoretical Physics, Institute for Advanced Study 2023

Ultra Quantum Matter. Summer school participant.

Boulder School for Condensed Matter and Materials Physics 2022

Non-Equilibrium Quantum Dynamics. Summer school participant.

Teaching Experience

Graduate Teaching Assistant, Cornell University

AEP 4230: Statistical Physics Fall 2026

Seminar instructor. Developed original coursework on statistical physics methods for neural networks.

PHYS 2214: Oscillations, Waves, and Quantum Physics Spring 2022

Seminar and laboratory instructor.

PHYS 1201: Why the Sky is Blue Fall 2021

Seminar instructor.

Mentoring

Volunteer Research Mentorship 2024-2025

Mentored high-school student Samyak Jain weekly for one year, from introductory quantum mechanics to a Qiskit project on entanglement dynamics in open quantum circuit ensembles. His project, *Entanglement Sudden Death in Interacting Quantum Circuits*, received third-place awards at the Massachusetts State and Region IV Science and Engineering Fairs (2025).

Graduate-school application advising 2023-2025

Advised two senior undergraduates on program fit and letter writers; substantially edited their statements of purpose. One received offers from UC Berkeley and Princeton (experimental condensed-matter physics); the other, from the University of Michigan (high-energy theory).

First-year cohort mentoring 2024

Met biweekly with first-year graduate students to discuss advisors, coursework, and graduate-student life.

Outreach

Moiré physics learning activity and teacher workshops 2022-2025

Co-developed with my research group a middle-school activity using stencils to measure moiré periods and connect moiré patterns to quantum materials; presented it at two Cornell-hosted teacher workshops.

Physics demonstrations for K-12 students 2025

Presented physics experiments to K-12 students at the Ithaca Sciencenter.